University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Final Exam
28th October, 2022
Time: 2 hours
Total marks: 85, Full marks: 80

- **Read the instructions carefully.**
- **You have 2hrs to complete the exam.**
- **Make sure that no pages or questions are missing.**
- **Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.**

Surname: ____________________

First name: ____________________

Student number: ____________________

Signature: ____________________

Do not write below this line:

Question:	1	2	3	4	5	Total
Marks:	15	25	15	15	15	85
Score:						

1. Let G be an arbitrary group and $a, b \in G$.

   (a) $\boxed{\text{10 marks}}$ Use induction to prove that $b(ab)^{n-1}a = (ba)^n$ for all $n \geq 1$.

   (b) $\boxed{\text{5 marks}}$ Assume that $(ab)^m = 1$ for some $m \geq 1$. Use (a) to show that $(ba)^m = 1$.

Solution: (a) For $n \geq 1$, let p_n be the statement $b(ab)^{n-1}a = (ba)^n$. It is clear that p_1 is true, since $b(ab)^{1-1}a = ba = (ba)^1$. Suppose now that p_k is true, that is, $b(ab)^{k-1}a = (ba)^k$ and let us show that p_{k+1} is also true. In fact:

$$b(ab)^k a = b(ab)(ab)^{k-1}a = (ba)\left(b(ab)^{k-1}a\right) \overset{\text{(I.H)}}{=} (ba)(ba)^k = (ba)^{k+1}.$$

By the Principle of Mathematical Induction, we obtain that p_n is true for all $n \geq 1$.

(b) Using that $(ab)^m = 1$ together with (a) we have that

$$(ba)^{m+1} = b(ab)^m a = ba.$$

Applying the Cancellation law, we obtain that $(ba)^m = 1$, as desired.

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

   (a) $\boxed{\text{5 marks}}$ Suppose that m and n are two positive integers. If $d > 1$ satisfies that $d = am + bn$ for some integers a and b, then $d = \gcd(m, n)$.

   (b) $\boxed{\text{5 marks}}$ The set of all permutations S_3 is an abelian group.

   (c) $\boxed{\text{5 marks}}$ Every abelian group is cyclic.

   (d) $\boxed{\text{5 marks}}$ Every cyclic group is abelian.

   (e) i. $\boxed{\text{4 marks}}$ Every nontrivial group G has at least two subgroups.

   ii. $\boxed{\text{1 mark}}$ What can you say about the least number of normal subgroups of G?

Solution: (a) **False:** The converse of Bézout's Lemma does not hold. Take for example, $m = 4$, $n = 8$. Then $\gcd(4, 8) = 4$ and 12 satisfies that $12 = 1 \cdot 4 + 1 \cdot 8$.

(b) **False:** The transpositions $\alpha = (1 \quad 2)$ and $\beta = (1 \quad 3)$ are in S_n and they do not commute. In fact:

$$\alpha\beta = (1 \quad 2)(1 \quad 3) = (1 \quad 3 \quad 2),$$
$$\beta\alpha = (1 \quad 3)(1 \quad 2) = (1 \quad 2 \quad 3).$$

(c) **False:** As seen in class the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic (all its elements have order 2) and it is abelian.

(d) **True:** Suppose that $G = \langle g \rangle$ is a cyclic group with generator G. Then any $a, b \in G$ are of the form $a = g^\ell$, $b = g^k$, for some integers ℓ, k. From here, we obtain that

$$ab = g^\ell g^k = g^{\ell+k} = g^{k+\ell} = g^k g^\ell = ba$$

An alternative reasoning: if G has order n, then $G \cong \mathbb{Z}_n$ and so G is abelian; if G is infinite, then $G \cong \mathbb{Z}$ and so G is abelian.

(e) i. **True:** $\{1\}$ and G are always subgroups of G.

ii. Now if G is not abelian, then $\{1\}$, G and $Z(G)$ are normal subgroups of G; if G is abelian, then $Z(G) = G$ and G has at least two normal subgroups, namely, $\{1\}$ and G.

3. Consider the additive cyclic group $\mathbb{Z}_{24}$.
   (a) $\boxed{\text{10 marks}}$ Find all the subgroups of $\mathbb{Z}_{24}$.
   (b) $\boxed{\text{5 marks}}$ Draw the lattice diagram of $\mathbb{Z}_{24}$.

Solution: (a) To determine the subgroups of $\mathbb{Z}_{24}$, we use the Fundamental Theorem of Finite Cyclic Groups, which tells us that $\mathbb{Z}_{24}$ has exactly 8 subgroups of orders the positive divisor of 24, namely, 1, 2, 4, 8, 3, 6, 12, 24.

To determine the unique subgroup of each of these orders, recall that the order of the subgroup $\langle \bar{d} \rangle = d\mathbb{Z}_{24}$ is the order of the element $\bar{d} = d\bar{1}$, which is given by the formula $\mathrm{o}\,(d\bar{1}) = 24/d$, provided $d|24$. Clearly, $\{\bar{0}\} = \langle \bar{0} \rangle$, and $\mathbb{Z}_{24}$ is the subgroup of $\mathbb{Z}_{24}$ of order 24.

$$\begin{aligned}
\text{Order 2: } 2 = \mathrm{o}\,(d\bar{1}) = \frac{24}{d} &\Rightarrow d = 12; && \text{the subgroup of order 2 is } \langle \overline{12} \rangle = 12\mathbb{Z}_{24};\\
\text{Order 4: } 4 = \mathrm{o}\,(d\bar{1}) = \frac{24}{d} &\Rightarrow d = 6; && \text{the subgroup of order 4 is } \langle \bar{6} \rangle = 6\mathbb{Z}_{24};\\
\text{Order 8: } 8 = \mathrm{o}\,(d\bar{1}) = \frac{24}{d} &\Rightarrow d = 3; && \text{the subgroup of order 8 is } \langle \bar{3} \rangle = 3\mathbb{Z}_{24};\\
\text{Order 3: } 3 = \mathrm{o}\,(d\bar{1}) = \frac{24}{d} &\Rightarrow d = 8; && \text{the subgroup of order 3 is } \langle \bar{8} \rangle = 8\mathbb{Z}_{24};\\
\text{Order 6: } 6 = \mathrm{o}\,(d\bar{1}) = \frac{24}{d} &\Rightarrow d = 4; && \text{the subgroup of order 6 is } \langle \bar{4} \rangle = 4\mathbb{Z}_{24};\\
\text{Order 12: } 12 = \mathrm{o}\,(d\bar{1}) = \frac{24}{d} &\Rightarrow d = 2; && \text{the subgroup of order 12 is } \langle \bar{2} \rangle = 2\mathbb{Z}_{24};
\end{aligned}$$

(b) To draw the lattice diagram we use that

$$\langle \bar{k} \rangle \subseteq \langle \bar{m} \rangle \Leftrightarrow m|k$$

$$
\begin{tikzcd}
 & \mathbb{Z}_{24} \arrow[dl, dash] \arrow[dr, dash] & \\
2\mathbb{Z}_{24} \arrow[d, dash] \arrow[drr, dash] & & 3\mathbb{Z}_{24} \arrow[d, dash] \\
4\mathbb{Z}_{24} \arrow[d, dash] \arrow[drr, dash] & & 6\mathbb{Z}_{24} \arrow[d, dash] \\
8\mathbb{Z}_{24} \arrow[dr, dash] & & 12\mathbb{Z}_{24} \arrow[dl, dash] \\
 & \{\bar{0}\} &
\end{tikzcd}
$$

4. (a) $\boxed{\text{5 marks}}$ State Lagrange's Theorem.

(b) Consider the following incomplete proof of **Lagrange's Theorem**:

> Suppose that $Ha_1, Ha_2, \ldots, Ha_k$ are the distinct cosets of H in G. Then $G =$ (A), which is a (B) union by the cosets properties. Next, using the cosets properties we have that $|Ha_i| =$ (C) for each i, so
>
> (D),
>
> concluding the proof.

With reference to this partial proof:

i. $\boxed{\text{2 marks}}$ Complete the sentence at (A).

ii. $\boxed{\text{1 mark}}$ Complete the sentence at (B).

iii. $\boxed{\text{2 marks}}$ Complete the sentence at (C).

iv. $\boxed{\text{5 marks}}$ Fill in the necessary steps at (D) to finish the proof.

Solution: (a) **Lagrange's Theorem.** Let H be a subgroup of a finite group G. Then the order G is a multiple of the order of H.

(b)

i. $G = Ha_1 \cup \ldots \cup Ha_k$.

ii. disjoint

iii. $|H|$

iv. $|G| = |Ha_1| + \ldots + |Ha_k| = k|H|$.

5. Let G be a group and H, K two subgroups of G such that K is normal in G.
   (a) $\boxed{\text{10 marks}}$ Prove that $H \cap K$ is a normal subgroup of H.
   (b) $\boxed{\text{5 marks}}$ Is $H \cap K$ a normal subgroup of K?

Solution: (a) Let us begin by noticing that $H \cap K$ is a subgroup of G, and so it is clearly a subgroup of H. To prove that it is normal in H, we use the Normality Test. For $h \in H$ we need to prove that

$$h(H \cap K)h^{-1} \subseteq H \cap K.$$

To do so, we take $a \in H \cap K$ and we show that $hah^{-1} \in H \ \cap K$. It is clear that $hah^{-1} \in H$, since $h, h^{-1}, a \in H$ because we are assuming that H is a subgroup of G. On the other hand, since $a \in K$ and K is normal in G, the Normality Test tells us that $hah^{-1} \in K$. From here we obtain that $hah^{-1} \in H \cap K$, as desired.

(b) Concerning to whether $H \cap K$ is normal in K, the answer is not necessarily. For instance, take $G = K = S_3$ and $H = \{\varepsilon, (1\ 2)\}$. As we proved in (b) H is a subgroup of S_3 which is not normal in S_3. We have that $H \cap K = H$ not normal in $K = S_3$.